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FilterFFPE: An Artifact Chimeric Read Filter to Improve SV Detection in FFPE Samples

Introduction

Technical ease and low storage costs make formalin-fixed paraffin-embedded (FFPE) tissue processing a common method for long term tissue specimen storage. However, formalin-fixation can result in fragmented, degraded, protein cross-linked DNA, introducing false positive results to next generation sequencing (NGS) data analysis[1]. Studies on FFPE artifacts have been focusing on noise in single nucleotide variant calling, with little attention paid to structural variation (SV). We found that FFPE samples are enriched with artifact chimeric reads that would result in a large number of false positive SV calls. Therefore we developed FilterFFPE, an R package to filter artifact chimeric reads in FFPE samples.

Methods

To remove the artifact chimeric reads but retain the chimeric reads of real SVs, FilterFFPE filters the reads from binary alignment map (BAM) file based on breakpoint information. Briefly, the filtering is based on the number of reads that share the same breakpoint and the presence of short complementary sequences around the breakpoint. To investigate the performance, we simulated series of FFPE samples with RSVSim[2] and our NGS read simulator SimFFPE[3] and used three structural variation detection tools for SV calling (Delly[4], Lumpy[5] and Manta[6]). Additionally, whole-exome sequencing (WES) data of seventeen real FFPE samples with matched fresh frozen samples were obtained from the European Nucleotide Archive (study accession: SPR044740 and PRJNA301548) and were analyzed for further evaluation of FilterFFPE (Fig.1).

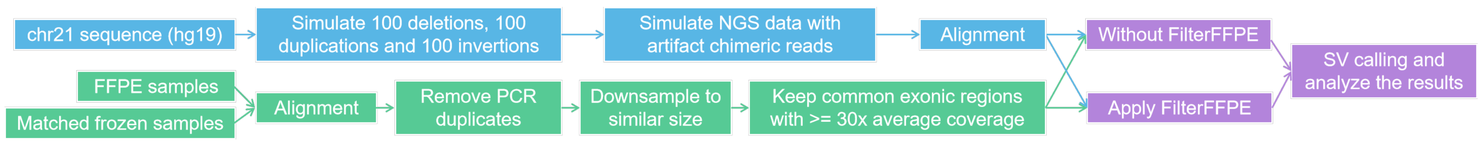

Fig. 1: Workflow to investigate the performance of FilterFFPE.

Results

Our results showed that FilterFFPE substantially improved the precision of SV detection in simulated samples with almost constant or even small increases in recall (Fig.2A). After applying FilterFFPE, the number of SV calls in real FFPE samples decreased remarkably, but remained stable in matched fresh frozen samples, and the number of shared SV calls between FFPE and matched fresh frozen samples remained almost unchanged. (Fig.2B).

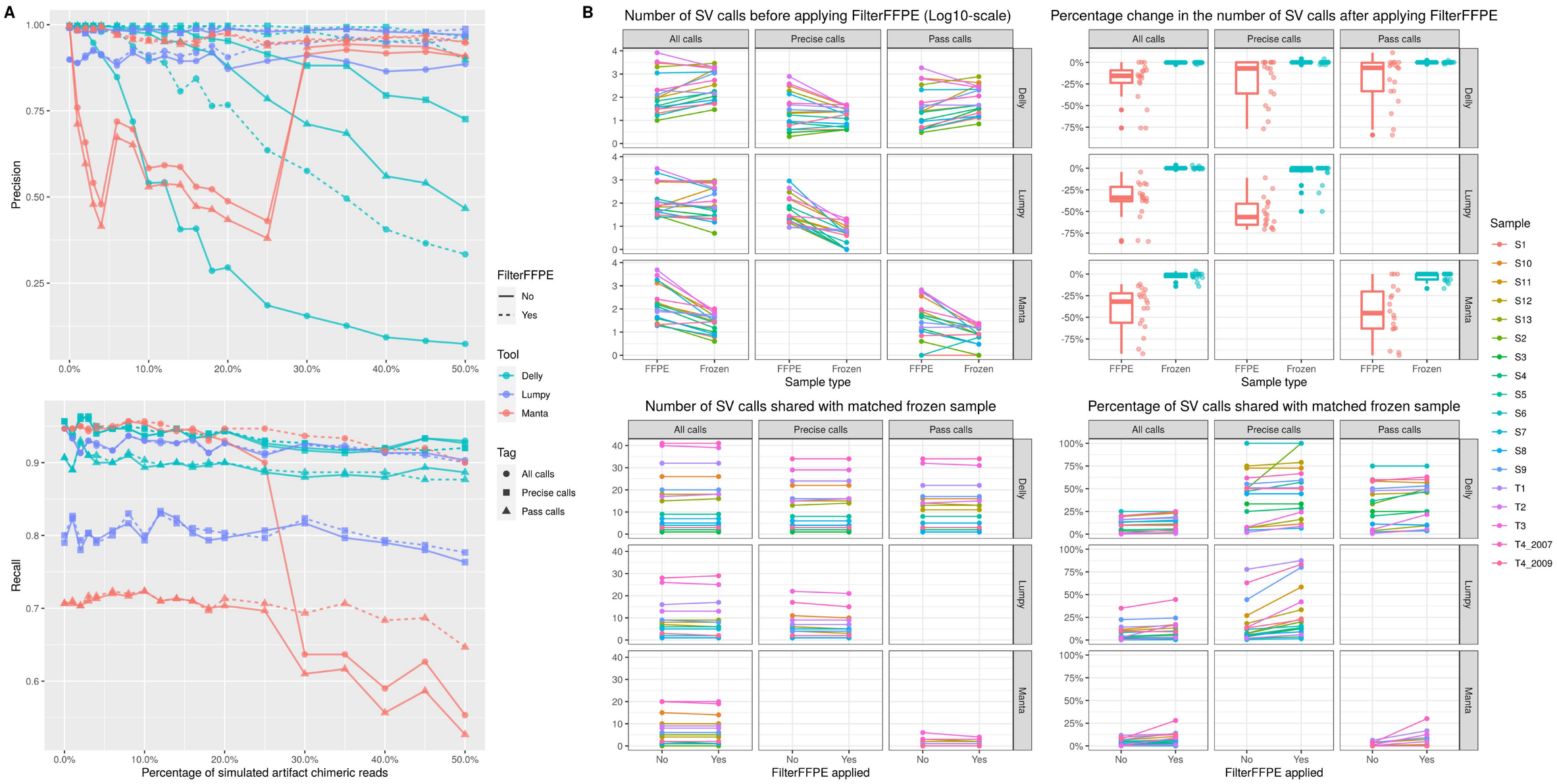

Fig. 2: Performance of FilterFFPE on improving SV calling in simulated samples (A) and in real FFPE samples (B). Pass and/or precise calls were labeled by each SV detection tool.

Conclusion

- FFPE samples have numerous false positive SV calls caused by artifact chimeric reads.
- FilterFFPE substantially reduces the number of false positive SV calls in FFPE samples by filtering artifact chimeric reads.
- For SV detection in FFPE samples, the combination of FilterFFPE and Delly (with precise label) shows favorable performance.

References

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